

Name: _____ Date: _____

Automation Badge 2 Vision Worksheet

Use this worksheet with Badge 2 Steps 10 and 12. Work out each prediction before running code, show units and frames, and bring your answers to the mentor review. You may use the linked lessons and a calculator. Keep your original prediction when you correct an answer.

1. Name the Frames

A frame has an origin and axes. In the names below, the first name is the reference frame and the second is the object or frame being located. Complete the table in your own words.

Name	Located Relative to Which Frame?	What Does It Describe?
fieldToRobot	_____	_____
robotToCamera	_____	_____
cameraToAprilTag	_____	_____
cameraToRobot	_____	_____

In the Robot frame, +X points forward, +Y points left, and +Z points up. In a top view, positive rotation is counterclockwise. For this worksheet, draw Field +X to the right and Field +Y upward on the paper.

A. Sketch a robot at Field position (2 m, 1 m), heading +90°. Label both Field axes and the robot's forward and left directions. Put a camera 0.5 m directly forward of the robot origin.

B. Write the camera's Field position. Explain why its 0.5 m forward offset does or does not add to Field X.

2. Apply and Reverse Transforms

Use 2D poses written as (x in meters, y in meters, heading in degrees). A transform's translation uses the axes of its starting frame. Sketch the directions before calculating.

A. The robot pose is (2, 1, +90). The camera mount transform is `robotToCamera = (0.5, 0, 0)`. Predict `fieldToRobot.plus(robotToCamera)`. Then repeat with the robot heading at 0°. The mount stays the same.

Robot Heading	Camera Field X	Camera Field Y	Camera Heading
+90°	_____	_____	_____
0°	_____	_____	_____

Explain which translation changed and why.

B. Consider a different mount: `robotToCamera = (0.5, 0, +90)`. Find `cameraToRobot`. Express its translation in Camera axes and give its rotation. Explain why negating X, Y, and the angle independently is not enough here.

C. A 3D vision calculation knows `fieldToAprilTag`, `cameraToAprilTag`, and `robotToCamera`. Fill the blanks to recover `fieldToRobot`. Each blank takes one supplied name, with `.inverse()` if needed.

```
fieldToAprilTag.plus(_____)
    .plus(_____)
```

Write the frame chain and cross out the matching inner names. Explain what the result of the first operation locates.

3. Calculate a Vision Residual

A translation residual compares two positions in the same frame and units. Here both positions are in the Field frame, in meters. Use observed minus reference. Its magnitude is a distance: the square root of $(\Delta x^2 + \Delta y^2)$. Heading is not part of this translation residual.

Case	Reference (x, y)	Observed (x, y)	Δx	Δy	Magnitude
A	(2.0, 1.0)	(2.3, 1.4)	_____	_____	_____
B	(2.0, 1.0)	(1.7, 0.6)	_____	_____	_____
C	(2.0, 1.0)	(2.0, 1.0)	_____	_____	_____

A. Show your calculation for Case A, including units.

B. Compare Cases A and B. What does the magnitude tell you, and what information about direction does it leave out?

C. Josh compares a camera pose in Field coordinates with a displacement in Robot coordinates. Both use meters. Explain what Josh needs to establish before subtracting them.

D. The camera captured its pose before the robot's current estimate was calculated. The robot moved during that interval. Explain why a large residual alone does not prove that the camera pose is wrong. Name one time-related value to inspect in the log.

4. Decide Which Observations Pass

Use these exercise settings: single-tag ambiguity must be **less than 0.20**, and the single-tag residual must be **less than 0.75 m**. These are practice values, do not use these for robot settings. Under the Badge 2 policy, observations using two or more tags pass both of those predicates automatically. Other predicates still apply.

Check in this order: `hasTags`, `ambiguityOk`, `onField`, `onFloor`, `timestampOk`, `residualOk`.

Record **accepted** or the **first failed predicate**. Field, floor, and timestamp checks pass unless a row says otherwise.

Case	Tags	Ambiguity	Residual	Other Failure	Decision
A	1	0.10	0.50 m	None	_____
B	1	0.20	0.50 m	None	_____
C	1	0.10	0.75 m	None	_____
D	2	Not used	1.20 m	None	_____
E	2	Not used	0.20 m	onField	_____
F	1	0.30	1.20 m	onFloor	_____
G	1	0.10	0.20 m	timestampOk	_____

A. Write the Boolean expression for acceptance using all six predicate names and Java operators.

B. For Case F, list every failed predicate, then explain why the recorded reason contains only one name.

C. Does passing every predicate prove that an observation is correct? Explain the separate role of uncertainty when an accepted observation reaches the pose estimator.

5. Track Each Camera and Check the Result

The estimator clock reads 12.00 s, with odometry history covering 10.50 s through 12.00 s. Initially, the front camera's last processed timestamp is 11.70 s and the rear camera's is 11.50 s. Process rows in order. A timestamp must be inside that history and newer than the last processed time for **that camera**.

For this exercise, advance a camera's last processed time when a newer frame is within history, even if a geometry check rejects it. All other checks pass except the one stated. Record accepted or the first failed predicate, then update both times.

Camera	Capture Time	Geometry Check	Decision	Front Last	Rear Last
Front	11.80 s	Pass	_____	_____	_____
Rear	11.60 s	Pass	_____	_____	_____
Front	11.80 s	Pass	_____	_____	_____
Front	11.90 s	onField fails	_____	_____	_____
Front	11.85 s	Pass	_____	_____	_____
Rear	10.40 s	Pass	_____	_____	_____

A. Explain why one shared last-processed timestamp would mishandle the second row. State why the capture time and age check need the same clock.

B. A camera result contains no estimated pose. What should be offered to the drivetrain for that result? Explain why inventing a zero pose would be misleading.

C. Choose one result from Sections 2–4 to check in your Badge 2 project. Record your prediction first, then the actual result and the test or log location. Explain any difference to the mentor.

Prediction: _____

Actual Result: _____

Test or Log: _____

Explanation: _____

References: [Step 10](#), [Step 12](#), [WPILib Coordinate System](#), and [WPILib Transformations](#).